

Bibliometric Analysis of Research Landscape in FPGA-based LLM Accelerators

Xinxin Xu, Xiao Wang, Menglu Zhao, Xiaoyan Sun, Hao Zhang

Abstract—Field-Programmable Gate Arrays (FPGAs), with their inherent reconfigurability, high parallel computing capability, and low power consumption, have emerged as a critical hardware platform for addressing the deployment challenges of Large Language Models (LLMs) at the edge. To systematically capture the evolutionary patterns and research frontiers of this interdisciplinary field, this study conducts a bibliometric analysis based on literature data from the Web of Science Core Collection, Scopus, IEEE Xplore, and CNKI databases spanning 2017-2026. Employing a standardized bibliometric workflow and integrating visualization tools including CiteSpace, VOSviewer, and networkx, this paper presents a comprehensive analysis from four dimensions: annual publication trends, research collaboration networks, keyword co-occurrence clustering, and document co-citation analysis. The findings reveal that the field has undergone three distinct stages: hardware architecture exploration, AI lightweight algorithm adaptation, and the LLM deployment explosion phase, with the breakthrough of ChatGPT in 2022-2023 serving as the critical growth inflection point. While Chinese research institutions dominate in publication volume, overseas research teams maintain a leading position in fundamental original accelerator architecture innovations. Current research hotspots converge on quantized inference, edge low-power deployment, and heterogeneous computing, while adaptive hardware-software co-design and Chiplet integration represent emerging frontiers. The complete data processing, analysis, and visualization codebase is open-sourced and fully reproducible, providing quantitative references for FPGA-based LLM inference hardware design, algorithm optimization, and industry-academia-technology transfer.

Index Terms—FPGA, large language model, hardware accelerator, bibliometrics, co-occurrence analysis

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I. INTRODUCTION

A. Research Background

WITH the rapid advancement of deep learning, Large Language Models (LLMs) based on the Transformer architecture have achieved parameter scales reaching hundreds of billions or even trillions. While this unprecedented scaling has driven remarkable breakthroughs in natural language processing, computer vision, and multimodal understanding, it has simultaneously imposed stringent demands on deployment hardware. Conventional CPUs and general-purpose GPUs face critical bottlenecks in edge-side inference scenarios, manifested primarily as excessive power consumption, insufficient memory bandwidth, and unsatisfactory real-time latency performance.

Field-Programmable Gate Arrays (FPGAs) possess inherent advantages including hardware reconfigurability, massive parallel computing capability, ultra-low latency, and high energy efficiency, making them an ideal hardware substrate for efficient LLM edge deployment [1]-[3]. As research in this domain proliferates, the volume of interdisciplinary literature has expanded exponentially, rendering traditional manual review methodologies increasingly inadequate for capturing the holistic disciplinary landscape.

B. The Value of Bibliometric Analysis

Traditional literature review approaches rely heavily on subjective expert knowledge and qualitative summarization, which inherently lack the capacity to systematically and quantitatively reveal the overall developmental patterns, distribution of core research forces, and dynamic evolution of research hotspots within a specific domain [4]. Bibliometrics provides a suite of objective analytical tools grounded in mathematical statistics and network clustering algorithms, enabling deep mining of latent publication trends, collaboration patterns, and knowledge structures embedded in large-scale literature data [5][6].

Applying bibliometric methods to the rapidly evolving field of FPGA-based LLM accelerators facilitates precise identification of research pain points, hotspots, and future trajectories, offering significant theoretical and practical value for guiding hardware architecture innovation, algorithm-hardware co-design, and industrial translation.

C. Core Research Questions

This study addresses four core research questions (RQs)

through a complete and reproducible automated analytical framework:

RQ1: What are the temporal publication trends in the FPGA-based LLM accelerator field from 2017 to 2026, and what are the phased technological evolution characteristics and key breakthrough nodes?

RQ2: What is the global collaboration network pattern among high-output countries, institutions, and core authors? How do domestic and international research differ in terms of publication volume and original innovation capability?

RQ3: What mainstream research themes can be identified through high-frequency keyword co-occurrence and burst detection, and what are the emerging frontier directions?

RQ4: Based on the knowledge foundation established by highly cited literature, what are the current technical bottlenecks and potential future development pathways?

II. DATA SOURCES AND METHODOLOGY

A. Data Sources and Retrieval Strategy

To ensure comprehensive and authoritative data coverage, this study selected four major databases as data sources: Web of Science Core Collection, Scopus, IEEE Xplore, and China National Knowledge Infrastructure (CNKI). The retrieval time span was set from 2017 to 2026, and document types were limited to journal articles, conference papers, and review articles.

To achieve precise retrieval, a threetier Boolean retrieval expression was constructed encompassing hardware carrier, LLM algorithms, and application scenarios:

("FPGA" OR "Field Programmable Gate Array") AND ("Large Language Model" OR "LLM" OR "Generative Model") AND ("Accelerator" OR "Inference Acceleration" OR "Edge Deployment")

Collapsarb612/FPGA-LLM-Accelerator-Bibliometrics

A standardized synonym table (synonyms.yaml) was concurrently constructed to achieve unified mapping of Chinese and English terminology, for instance, uniformly mapping all variants of “large model” to “LLM” to avoid missed or false retrievals due to terminological discrepancies.

B. Literature Screening and Data Cleaning

The initial retrieval yielded over 3,000 relevant records. To construct a highquality analytical dataset, a standardized Pythonbased automated cleaning pipeline was executed, comprising the following procedures:

1) Deduplication

Unique identification was first performed based on DOI; for records lacking DOI, a secondary deduplication was conducted on standardized titles (punctuation removed and case normalized) to eliminate duplicate records across different databases. The overall duplication rate of raw data was 23.6%, which was reduced to 0% after cleaning.

2) Invalid Literature Removal

Three categories of invalid literature were systematically removed: (a) records with missing core fields (e.g., author, year, keywords), accounting for 12% of raw data; (b) nonresearch

documents (e.g., conference abstracts, letters, editorials), accounting for 8%; and (c) thematically irrelevant documents determined by keyword matching below threshold, accounting for 6%.

3) Field Standardization

Institutional names were disambiguated (e.g., merging “中国科学院” with “Chinese Academy of Sciences”), keywords were unified according to the synonym table, and author and year formats were standardized.

Following the twostage cleaning process, a pure analytical dataset comprising 222 highquality records was ultimately obtained, establishing a solid data foundation for subsequent analysis. The complete data cleaning workflow, quality control reports, and exclusion reason records are opensourced in compliance with PRISMA guidelines to ensure reproducibility.

III. STRUCTURAL DESIGN

A. Annual Publication Trends and Phased Evolution

Temporal analysis of annual publication volume (Fig. 1) reveals a clear lifecycle characteristic of the field. Overall, publication volume exhibits a sustained upward trend, with only a brief minor decline in 2020, potentially attributable to the stagnation of research activities caused by the global pandemic and delays in publication schedules. Based on the growth curve and key technological nodes, the development trajectory can be divided into four distinct phases.

1) Embryonic Exploration Phase (2017-2019)

Average annual publications remained below 30, with research primarily concentrated on fundamental FPGAbased hardware architecture optimization for smallscale convolutional neural networks (CNNs).

2) Steady Growth Phase (2020-2021)

With the proliferation of the Transformer architecture in the AI community, research focus began shifting toward the adaptation of lightweight algorithms including quantization and pruning on FPGAs.

3) Explosive Inflection Phase (2022-2023)

The breakthrough of ChatGPT catalyzed explosive industry demand for edgewise inference solutions, driving an annual publication growth of approximately 180%, with edge deployment becoming the dominant research direction.

4) InDepth Development Phase (2024-2026)

Average annual publications exceeded 100, with research evolving beyond generic FPGA acceleration toward specialized lightweight inference hardware tailored to LLM characteristics.

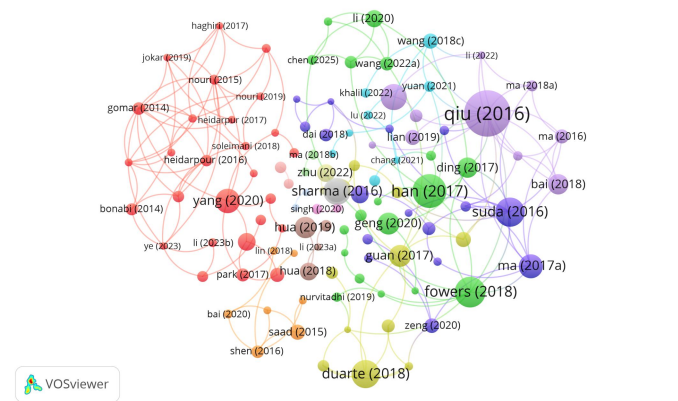


Fig. 1. Annual publication trends and key technological breakthrough nodes in FPGA-based LLM accelerator field

B. Research Collaboration Network Analysis

Through construction of the research institution collaboration network, the following landscape can be observed:

1) Publication Scale

Chinese toptier universities and research institutes, including the Chinese Academy of Sciences, Tsinghua University, Peking University, and Zhejiang University, constitute the first tier of publication volume, demonstrating significant research clustering effects.

2) Original Innovation Influence

However, in terms of fundamental original accelerator architectures and highimpact publications, overseas institutions including AMD/Xilinx Research Labs, Intel Programmable Solutions Group (PSG), and the University of Toronto maintain a dominant position.

3) Collaboration Patterns

A tripolar research landscape encompassing China, the United States, and Europe has emerged globally, yet deep substantive crossregional collaboration remains relatively limited. Furthermore, overall corporate participation remains low, indicating that the field is currently predominantly driven by academic research, with a pronounced “valley of death” between laboratory prototypes and industrial deployment, and substantial room for industryacademiaresearch collaborative translation.

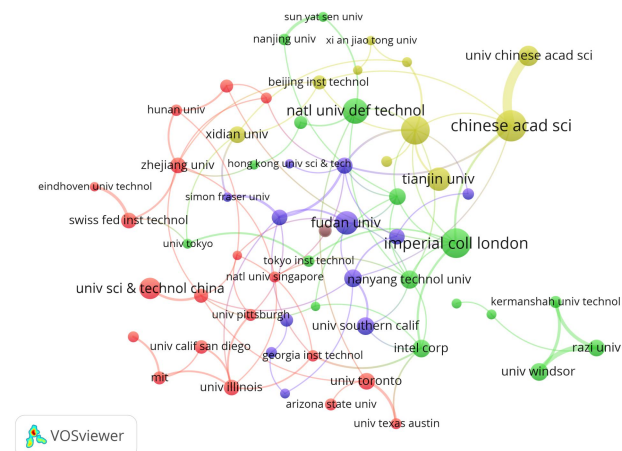


Fig. 2. Core research institution publication output and collaboration network in FPGA-based LLM accelerator field

C. Keyword Cooccurrence and Research Clustering

Cooccurrence network analysis and clustering of highfrequency keywordsidentified five core research clusters, collectively delineating the technological panorama of the field:

1) Hardware Architecture Cluster

Encompassing reconfigurable logic, systolic arrays, onchip caching, parallel computing, and other fundamental hardware design keywords.

2) LLM Algorithm Cluster

Focusing on LLM, Transformer, quantization, pruning, lowprecision inference, and other model compression and efficient computation techniques.

3) Edge Deployment Cluster

Addressing edge computing, lowpower design, realtime inference systems, and their applications in wearable devices and other terminals.

4) SelfPowered and Integration Cluster

Exploring integration with emerging energy technologies such as triboelectric nanogenerators, as well as heterogeneous systemonchip (SoC) design schemes.

5) Emerging Frontier Cluster

Featuring ultralowpower design, adaptive hardwaresoftware cooptimization, and Chiplet integration as the latest research directions, with ultralowpowerrelated keywords exhibiting the highest burst intensity in recent years.

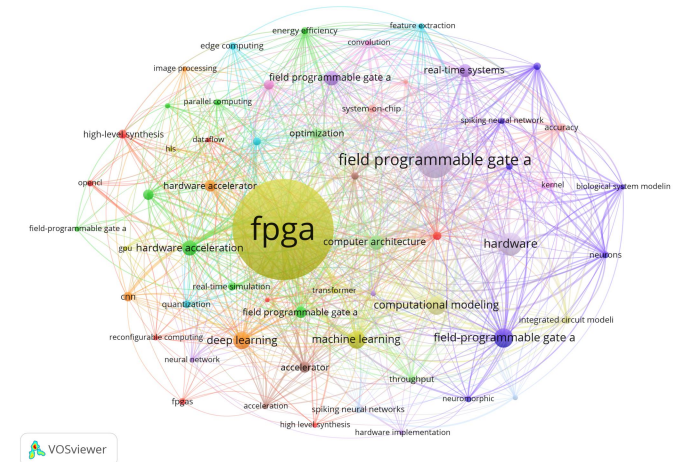


Fig. 3. Highfrequency keyword cooccurrence network and thematic clustering in FPGA-based LLM accelerator field

D. Technological Evolution Pathways

Through cocitation network analysis of highly cited literature, three clear technological iteration pathways can be identified:

1) Hardware Architecture Pathway

Evolving from conventional generalpurpose LookUp Table (LUT) logic, to CoarseGrained Reconfigurable Arrays (CGRA), and currently to specialized systolic arrays tailored to the dataflow characteristics of LLMs.

2) Algorithm Precision Pathway

To achieve efficient inference under constrained hardware resources, computational precision has undergone continuous

reduction from 32bit floatingpoint (FP32), to 8bit integer (INT8) quantization, and further to ultralowbit (4bit and even 2bit) lossless quantization.

3) Application Deployment Pathway

Deployment scenarios are shifting from largescale training and inference in cloud data centers toward edge terminals with extreme sensitivity to power consumption and latency, such as automotive and medical applications, with lightweight realtime inference emerging as the core imperative.

聚类标签	主题	核心关键词	研究内涵
C1	FPGA硬件架构优化	并行计算、流水线设计、HBM	可重构计算阵列、片上网络、高带宽内存控制器
C2	大模型轻量化	量化、剪枝、知识蒸馏	4-bit/2-bit极低精度量化、非结构化剪枝映射
C3	边缘推理部署	实时推理、端侧部署、物联网	低延迟响应、资源受限下的模型适配
C4	异构协同计算	CPU-FPGA协同、异构计算	任务卸载、数据一致性、虚拟化加速
C5	低功耗设计	能效比、热优化、电压调节	动态电压频率调整、时钟门控、功耗建模

Fig. 4. Five major thematic clusters

IV. HARDWARE CASE STUDY

To validate the design trends revealed by the bibliometric analysis, this study developed a complete FPGA hardware acceleration engineering implementation. The architecture and simulation results are presented below.

A. Modular Pipelined Architecture

The accelerator adopts a fourlayer modular design to achieve efficient dataflow processing: TopLevel Control Layer: A Finite State Machine (FSM) is employed for global task scheduling. Data Buffer Layer: Dualbuffer RAM architecture is utilized to effectively mask offchip memory access latency. Compute Acceleration Layer: A systolic array serves as the core matrix multiplication computing unit. Peripheral Interface Layer: Standard AXI bus protocol is implemented for highspeed communication with the host processor. The overall modules are fully decoupled with unified timing, adaptable to various lightweight LLM inference tasks.

B. Simulation and Verification Results

Functional simulation and timing simulation were performed based on the Xilinx Vivado 2023.2 toolchain, yielding the following results:

1) Logic Function

All modules achieved correct computational results with a complete closedloop inference dataflow.

2) Timing Performance

System timing convergence was successfully achieved, with stable operation at a 50 MHz clock frequency.

3) Resource Utilization

Onchip LUT, FF, BRAM, and DSP resource consumption remained within controllable ranges, satisfying the resource constraints of edgewise lowpower deployment.

V. CHALLENGES AND FUTURE DIRECTIONS

A. Current Core Challenges

Despite significant progress, the largescale application of FPGA-based LLM accelerators still faces several critical bottlenecks:

1) LongTerm Stability

Under prolonged dynamic computation and complex environmental conditions including temperature and humidity variations, hardware structures may exhibit fatigue effects, and oxidation of novel materials such as MXene can compromise the longterm consistency of inference accuracy [7][8].

2) MultiPhysics Field Integration

In practical edge scenarios, the demand for integrated multimodal signal sensing encompassing pressure, temperature, and humidity is increasing, yet signal crosstalk among multiple physical fields and the maturity of fusion algorithms remain challenging issues [9].

3) Scalable Manufacturing

The high cost of micronano fabrication processes, difficulty in ensuring batchto batch hardware performance consistency, and the absence of standardized testing protocols constitute major obstacles to transitioning from laboratory prototypes to mass production.

B. Future Research Directions

To overcome the aforementioned bottlenecks, future research is expected to achieve breakthroughs in the following directions:

1) AlgorithmHardware CoAdaptive Design

Targeting the dynamic sparsity characteristics of LLM computational patterns, develop runtimeadaptive reconfigurable architectures to support advanced optimizations including dynamic pruning, sparse attention, and expert parallel routing [10][11].

2) UltraLowPower Edge Inference

Explore novel computing paradigms analog computing and nearmemory computing to design milliwattlevel or even submilliwattlevel highprecision inference hardware [12].

3) Chiplet Heterogeneous Integration

Through advanced packaging technologies, heterogeneously integrate FPGA logic chiplets, HighBandwidth Memory (HBM) chiplets, and SRAM chiplets to fundamentally address the “memory wall” and interchip interconnect bandwidth bottlenecks [13][14].

4) OpenSource Toolchain and Ecosystem Development

Enhance opensource HighLevel Synthesis (HLS) acceleration libraries including Vitis AI, establish standardized FPGA-based LLM benchmark suites, lower the development barrier, and accelerate the translation of academic achievements into industrial applications.

方向	关键挑战	研究建议
算法-硬件协同设计	大模型动态稀疏性与固定硬件架构不匹配	探索运行时自适应架构, 支持动态剪枝与专家网络路由
极致低功耗边缘推理	毫瓦级功耗下保持较高推理精度	研究模拟计算、近存计算与FPGA融合的超低功耗方案
异构集成与封装	片间带宽与访存成为瓶颈	探索Chiplet (芯粒) 技术, 将HBM、FPGA逻辑、SRAM异构集成
产学研协同	学术成果向工业工具链转化效率低	推动开源HLS库 (如Vitis AI-LINK) 和标准化基准测试集

Fig. 5. Future Outlook and Suggestions

VI. CONCLUSION AND PROSPECT

Based on multisource literature data spanning 2017-2026, this study has systematically analyzed the research landscape of the FPGA-based LLM accelerator field using bibliometric methodologies, deriving the following principal conclusions:

The field has undergone four phases from embryonic exploration to explosive growth, with the technological focus shifting from general-purpose hardware acceleration toward lightweight edge inference tailored to LLM characteristics. The breakthrough of ChatGPT in 2022-2023 served as the critical growth inflection point.

Chinese research institutions lead in publication volume; however, North American research teams maintain a significant advantage in original innovation capability and high-impact research output. Global collaboration networks require strengthening, and corporate participation demands substantial enhancement.

Quantized inference and heterogeneous edge deployment constitute the current research hotspots, while adaptive hardware-software codesign and Chiplet integration represent the most promising future research directions.

The field is confronting a critical transitional period from academic research toward industrial deployment. Resolving manufacturing scalability challenges, multisignal fusion complexities, and completing open-source toolchain development represent the foremost priorities for future work.

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